

Objects that depend on other objects and can modify the states of other objects are called tightly coupled. It creates conditions where modifying the code of one object also requires changing the code of other objects. The reuse of code is difficult in tight coupling because we cannot separate the code.

Since using loose coupling is always a good habit.

29) Name the operators that cannot be overload.

1. Scope Resolution Operator (::)
 2. Ternary Operator (? :)
 3. Member Access or Dot Operator (.)
 4. Pointer to Member Operator (.*)
 5. sizeof operator
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30) What is the difference between new and override?

The new modifier instructs the compiler to use the new implementation instead of the base class function. Whereas, Override modifier helps to override the base class function.

virtual: indicates that a method may be overridden by an inheritor

override: Overrides the functionality of a virtual method in a base class, providing different functionality.

new: Hides the original method (which doesn't have to be virtual), providing different functionality. This should only be used where it is absolutely necessary.

When you hide a method, you can still access the original method by upcasting to the base class. This is useful in some scenarios, but dangerous.

31) Explain overloading and overriding with example?

Overloading

Overloading is a concept in OOP when two or more methods in a class with the same name but the method signature is different. It is also known as